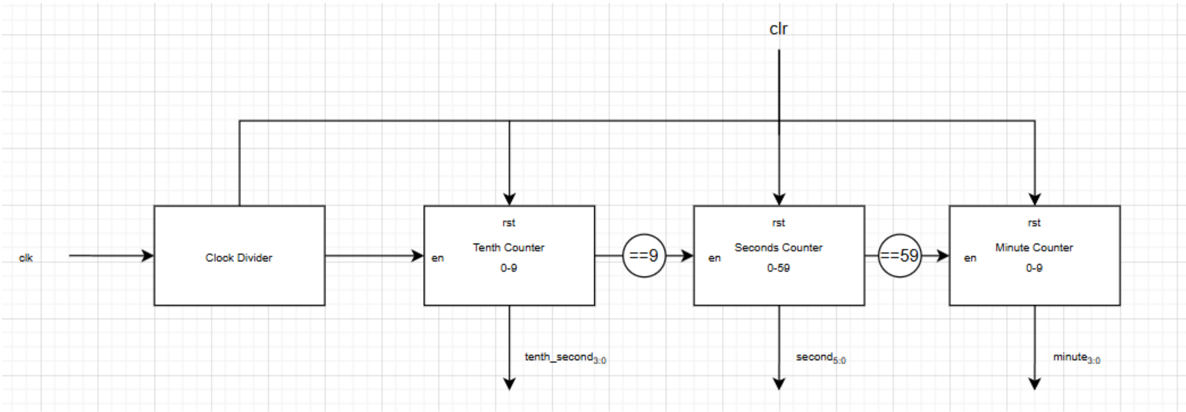
Task 13

https://edaplayground.com/x/gCvh



Block Diagram

Internal Counter Timing Calculation

The stopwatch uses a 100 MHz system clock as its input.

Step 1: Calculate the clock period

Clock Period = 1 / Clock Frequency

= 1 / 100,000,000

= 10 ns

Step 2: Calculate the required counter value

The stopwatch must generate one timing tick every 0.1 seconds (100 ms).

Number of clock cycles required:

= 0.1 s / 10 ns

= 0.1 / (10×10^{-9})

= 10,000,000 clock cycles

Therefore, the internal counter must count 10,000,000 clock cycles before generating one tick.

Step 3: Calculate the output frequency

Output Frequency = 1 / Tick Period

= 1 / 0.1

= 10 Hz

Thus, the internal counter generates a 10 Hz enable pulse.

Step 4: Calculate the pulse width

In the design, the signal clk_out becomes high only when the counter value is zero. Therefore, it remains high for one clock cycle.

Pulse Width = 1 × Clock Period

= 10 ns

Step 5: Calculate the duty cycle

Duty Cycle = (Pulse Width / Tick Period) × 100%

= (10 ns / 0.1 s) × 100%

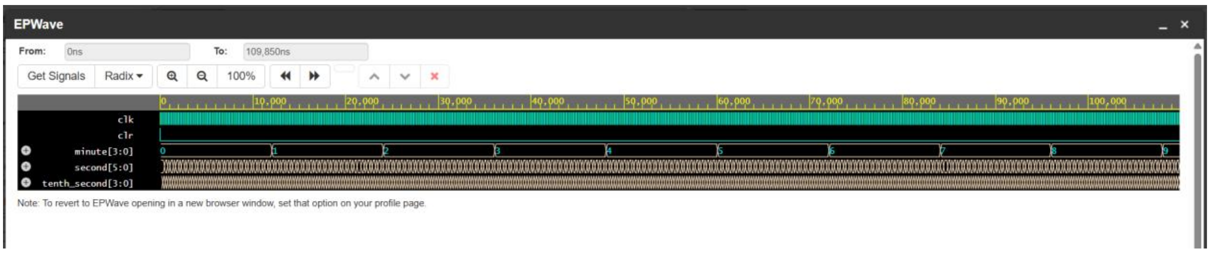
= $(10 \times 10^{-9} / 0.1) \times 100\%$

= 0.00001%

Summary

Parameter	**Value**
Input clock frequency	100 MHz
Input clock period	10 ns
Counter value	10,000,000 cycles
Tick period	0.1 s (100 ms)
Output (tick) frequency	10 Hz
Pulse width	10 ns
Duty cycle	0.00001%

The internal counter divides the 100 MHz system clock to produce a 10 Hz enable pulse. This pulse is high for only one clock cycle (10 ns), resulting in a duty cycle of 0.00001%. The enable pulse is used to increment the tenths-of-a-second counter while all registers continue to operate using the original 100 MHz clock.



Waveform Screenshot